Reinventing the Sacred: Back to the Future of Early-Day Internet Pioneers

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***Abstract*—This is not a paper for the timid, but it will reward the courageous and open-minded. In the coming Age of AI, nothing could be more dangerous for the future of AI entrusted on us than a closed mind. We have to learn in a hurry or we miss an incredible opportunity that presents itself by revisiting the "untold history of AI" of early-day Internet pioneers and their (almost) forgotten vision of a human-centered AI (HAI). After briefly reviewing Licklider's AI vision of facilitating previously unimaginable forms of intelligent human activity that seem at times to behave like a "global brain," this paper focuses on the future *6G world brain*, made feasible thru enabling 6G technologies, most notably AI embodiment, agentic AI, and integrated sensing and communications (ISAC), which together create a global action-perception loop that is typically found in living intelligent systems. We show that by advancing active inference, a cornerstone of the emerging field of *NeuroAI*, novel biologically informed artificial general intelligence (AGI) models arise, which harness the increased neuroplasticity of the 6G world brain for us humans to penetrate deeper and deeper into cognitive space, as originally envisioned by early-day Internet pioneers.**

***Index Terms*—6G world brain, active inference, AI embodiment, enactive AI, human-centered AI (HAI), quantum biology.**

I. Introduction

A few years ago, we were arguing in [1] that the emerging 5G/6G Tactile Internet premised on embodied human-to-robot interaction may be the harbinger of *human augmentation* (rather than human automation, i.e., replacement of humans by machines) envisioned by contemporary and early-day Internet pioneers. Specifically, we were quoting Garry Kasparov who famously stated in his book titled "Deep Thinking: Where Machine Intelligence Ends and Human Creativity Begins" that weak human + machine + better process is superior to strong human + machine + inferior process. Thus, a clever process beats superior knowledge and superior technology. His observation received interest by Google and other Silicon Valley companies and shifted the research focus from artificial intelligence (AI) to *intelligence amplification (IA)*.

Interestingly, this IA approach is fully in line with the original vision of early-day Internet pioneers. Back in 1962, Douglas C. Engelbart developed a detailed, though rudimentary, conceptual framework with process hierarchies for augmenting the human intellect by increasing via on-line assistance human capabilities to drive solutions to complex problems that before seemed insoluble [2]. Earlier, in 1960, Joseph C. R. Licklider envisioned man-computer symbiosis, a subclass of man-machine systems, to enable close interaction between man and computer in mutually beneficial cooperation [3]. Fast forwarding, in [4], Kevin Kelly argues that in terms of the Internet, nothing has happened yet. Among other technological forces that will shape our future, Kelly highlights that *cognifying*, i.e., embedding AI into an existing process or inert thing, will be hundreds of times more disruptive to our lives than the transformations gained by industrialization. According to Kelly, the future Internet will include the collective intelligence of all humans combined with the collective behavior of all machines, plus the *intelligence of nature*.

It is interesting to note that Yann LeCun, one of the "godfathers" of AI, has recently left Meta after being the founding director of the Fundamental AI Research (FAIR) lab and chief AI scientist for 12 years. In [5], LeCun explained that he is at odds with where AI is heading and that he wants to pursue research on advanced machine learning by *"replicating how a child or a baby animal learns,"* as opposed to current LLMs that are fed vast amounts of training data. This paper expands on our recent work in [6], where we have introduced *active inference* as an ideal methodology for developing more advanced AI systems by biomimicking the way living intelligent systems work, ranging from humans and animals to plants and other life forms.[1] Active inference is a biomimetic mathematical framework developed by the world's most frequently cited neuroscientist Karl Friston. It is premised on first principles of statistical physics found in self-organizing/evolving complex adaptive systems, whether natural, artificial, or hybrid cyborganic ones. The specific goal of this paper is to study the future *6G world brain*, where AI neurons are replaced with humans and active inference agents, and *how to enhance its neuroplasticity* to learn new patterns of IA by cracking open the door of human perception.

The remainder of the paper is structured as follows. Section II briefly reviews the "untold history of AI" between the

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[1]The title of this paper is inspired by the book *"Reinventing the Sacred: A New View of Science, Reason, and Religion"* by Stuart A. Kauffman, a worldwide renown theoretical biologist and complex systems researcher who studies the origin of life on Earth.

competing AI visions of Turing and Licklider, making the case that human capacity goes beyond computation. Section III echoes this viewpoint by elaborating on the importance of "true" intelligence today's digital machines don't have, paving the way to the emerging field of *NeuroAI* that aims at fusing neuroscience and AI for truly intelligent solutions by means of advanced biologically-informed AI models that don't lose sight of what makes natural intelligence so remarkable. Next, Section IV elaborates on active inference and describes our proposed advancements to foster the aforementioned research shift from AI to IA. Finally, Section V provides a research outlook on the future of active inference premised on *quantum biology* before concluding in Section VI.

II. Competing AI Visions: "Untold History of AI"

According to IEEE Spectrum [7], the history of AI is often told as the story of machines getting smarter over time, but what's lost is the human element in the narrative. What is often forgotten in this "untold history of AI" is that underlying the Internet was a radical vision for a future of human-machine symbiosis developed by J. C. R. Licklider, who had a background in psychology and was interested in how machines could amplify humanity's collective intelligence. Instead of having machines imitate human intellectual activities, Licklider's approach would shift the focus from competition (e.g., computer-versus-human chess matches) to collaboration between humans and machines in order to facilitate previously *unimaginable forms of intelligent activity* that seem at times to behave like a "global brain."

In the 1960s, the inaugural decade of AI research, early experiments in computational creativity have been informed by two competing visions for AI put forward by J. C. R. Licklider on one hand and Alan Turing on the other hand [8]. While Turing's automative vision of AI fostered the idea of computers replicating, automating, and finally replacing humans, Licklider's vision of a hybrid cybernetic organism (cyborg) AI maintained that there are certain genotypic differences in the capability between men and computers and that men can do readily well, though not rapidly, many things that are difficult or impossible for computers since *human capacity goes beyond computation*. Unlike Turing, Licklider's cyborg AI vision of human augmentation (in lieu of automation) for the exploration and discovery of new human behaviour is capable of extending human capacity rather than imitating pre-existing human activities that were previously impossible without the given computational tool of AI. Clearly, this Licklider-derived vision of AI helps us open our eyes to alternative conceptions of future AI and has been revisited under the new name of *human-centered AI (HAI)*, as exemplified by the Stanford Institute for HAI, among other leading AI research centers [9].

III. Artificial Intelligence vs. Natural Intelligence

In their recent book on "Artificial Intelligence versus Natural Intelligence" [10], physics Nobel Laureate Sir Roger Penrose *et al.* showed that the machines at our disposal today (e.g., computers or computer-controlled robots) are computational devices, maybe very sophisticated, but all essentially based on the ideal prototype of the algorithmic Turing Machine. They argue that intelligence, on the other hand, exhibits properties that escape simple computatbilitiy, i.e., intelligence is something more than mere computational ability. According to Penrose, "true" intelligence requires consciousness, something that our digital machines do not have, and never will. He goes on to say that consciousness is not a feature which is restricted to humanity, and he strongly believes it's true with animals, which can be extremely creative in a certain sense.

There is now a growing awareness among leading researchers that there is the risk of over-reliance on computational tools at the expense of biological insights, giving rise to the emerging field of *NeuroAI* [11]. Historically, NeuroAI has referred to how principles of neural architecture and function can inspire AI models. However, the dynamics of the neuroscience-AI relationship seems to have been reversed recently. While current NeuroAI focuses primarily on applying AI tools to neuroscience, the reverse direction can be equally transformative, leading to the development of *biologically informed AI models*. The next generation of NeuroAI researchers with both strong computer science skills and solid biological foundations will be crucial in preventing advanced AI research from losing sight of what makes natural intelligence so remarkable in the first place.

Towards this end, research initiatives have been launched such as the 2024 NeurIPS Workshop on "NeuroAI: Fusing Neuroscience and AI for Intelligent Solutions," which aim at examining the fusion of neuroscience and AI to unlock biologically informed AI models, including a panel discussion between two of the biggest names in AI today: Yoshua Bengio and Karl Friston [12]. While Bengio is most known for his pioneering work in deep learning – earning him the 2018 A. M. Turing Award, "the Nobel Prize of Computing," with Geoffrey Hinton and Yann LeCun – Friston is the world's most frequently cited neuroscientist and the father of *active inference* – considered the key to true AI (as well as Penrose's above mentioned "true" intelligence), as explained next.

IV. Active Inference - The Key to True AI: Less Artificial, More Intelligent

Active inference is a unified theory premised on first principles of statistical physics, which reconciles and unites embodied, enactive, and cybernetic theories of brain, mind, and life in general [13]. According to a widely read WIRED article, in the end, the acronym AI might actually not stand for artificial intelligence by active inference – it is considered *the key to true AI: Less artificial, more intelligent* [14].

In stark contrast to the monstrous scaling energy demands of today's AI systems (e.g., large language models like ChatGPT), active inference facilitates the most energy-efficient form of learning with no big data requirement necessary for training. This is due to the fact that active inference leads to a novel view of optimal behavior in terms of the so-called Hamiltonian principle of *least action* – a first principle

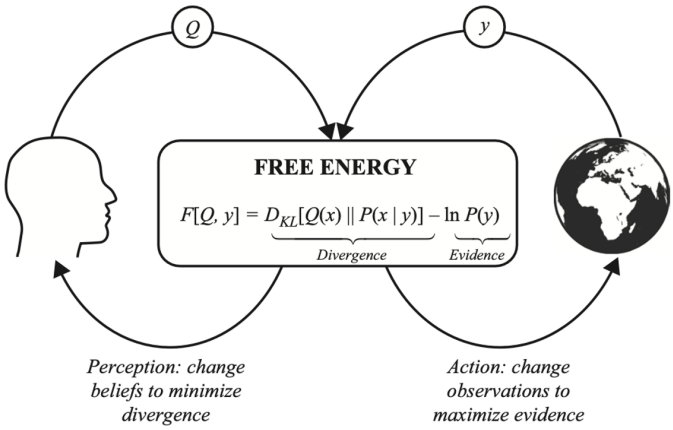

Fig. 1. The action-perception loop of embodied, enactive biological agents: Living organisms achieve lifelong learning by continuously minimizing the discrepancy (i.e., free energy) between belief Q (i.e., world model) and observation y (i.e., perception) of the real world. Biological agents use action to change observations to maximize evidence of their world model and use perception to change beliefs to minimize divergence from it [13].

that connects active inference to the domains of statistical physics, thermodynamics, and nonequilibrium steady states. The physics underlying active inference are pointing in the exact opposite direction of today's AI trajectory. From the point of view of the physicist, one should be looking for the *least data* instead of big data in order to resolve uncertainty and to learn more intelligently in doing so [15].

Active inference is an ideal methodology for developing advanced *artificial general intelligence (AGI)* systems by biomimicking the way natural intelligence works based on the action-perception loop of *embodied, enactive* biological agents (i.e., humans, animals, plants, etc.) and the so-called *free energy principle*, another first principle of statistical physics.

A. AGI via Action-Perception Loop & Free Energy Principle

Living organisms continuously learn by interacting with their physical environment via the action-perception loop, which naturally lends itself to *lifelong learning*, a capability that has been studied in robotics for decades [16], but is completely lacking in today's AI, e.g., generative AI (GenAI) such as ChatGPT. As illustrated in Fig. 1, the free energy principle (FEP) is essentially a mathematical formulation of how self-organizing/evolving adaptive systems avoid surprising states for the sake of their survival in the face of a constantly changing environment. They can do this by minimizing their free energy, which can be shown to be an upper bound on the surprise of sensory states. Minimizing free energy helps an active inference agent to find beliefs in its world model for which observations provide the most evidence, a process known as *self-evidencing*. The FEP explains how biological systems in general learn, adapt, and self-evolve via AGI capabilities such as lifelong learning and common sense (i.e., causal understanding of real world) that all humans naturallly have in common, as opposed to today's AI [17].

Note that active inference subsumes a number of established reinforcement learning (RL) formalisms for adaptive decision-making in unknown environments. However, unlike RL, active inference enables agents to acquire a causal world model thanks to the FEP (in lieu of the rather simplistic reward function in RL). It is important to note that this world model enables active inference agents to learn the causal structures of their worlds instead of merely correlations, thus avoiding the need for (re)training in different (changing) environments.

Formally, the free energy $F[Q, y]$ is given by

$$F[Q,y] = \underbrace{D_{KL}[Q(x) \mid\mid P(x|y)]}_{\text{Divergence}} + \underbrace{[-\ln P(y)]}_{\text{Surprise}} \quad (1)$$

$$= \underbrace{D_{KL}[Q(x) \mid\mid P(x|y)]}_{\text{Divergence}} - \underbrace{\ln P(y)}_{\text{Evidence}}, \quad (2)$$

where D_{KL} denotes the Kullback-Leibler (KL) divergence between the two distributions $Q(x)$ and $P(x|y)$ of an agent's belief x and observation y, respectively. The term $-\ln P(y)$, i.e., negative log model evidence, is called surprisal in information theory, or often surprise for short. Note that surprise is the inverse of evidence, i.e., minimizing surprise is equivalent to maximizing evidence, as done in active inference.

B. Embodied & Enactive AGI: Toward the 6G World Brain

Recently, Friston *et al.* laid out an AGI vision and R&D roadmap premised on active inference for the next decade and beyond [18]. While acknowledging neuroscience as a key inspiration for AI research, Friston argues that we must move beyond brains and embrace the active characteristics of natural intelligence in living organisms. He suggests that AGI will emerge from the interaction of intelligence networked into a *cyber-physical ecosystem of artificial and natural intelligence*, in which humans are integral participants – what he calls "shared intelligence." Key to such an ecosystem will be the interaction between artificial and natural intelligence, giving rise to *human-AI interaction*, a nascent research field of utmost importance for future communication networks such as 6G and the 3D spatial Internet commonly referred to as the *Metaverse*, the anticipated successor of today's mobile Internet.

It is also worthwhile mentioning that a recent research outlook on the evolution of future AGI-native networks in the beyond 6G era envisions a new breed of AGI-native 6G networks operating with an *AGI-native telecom brain*, capitalizing on emerging 6G key enabling technologies such as integrated sensing and communications (ISAC) as well as AI embodiment and agentic AI [19]. While ISAC will be instrumental in realizing a distributed sensing environment, AI embodiment and agentic AI will allow AI agents to transcend their purely computational nature and enable them to interact with their surrounding physical environment via sensory grounding through 6G. Note that AI embodiment and agentic AI in combination with ISAC will form a *global action-perception loop* for enabling active inference.

Figure 2 depicts our proposed approach to realize the 6G world brain in the emerging Metaverse, paying particular

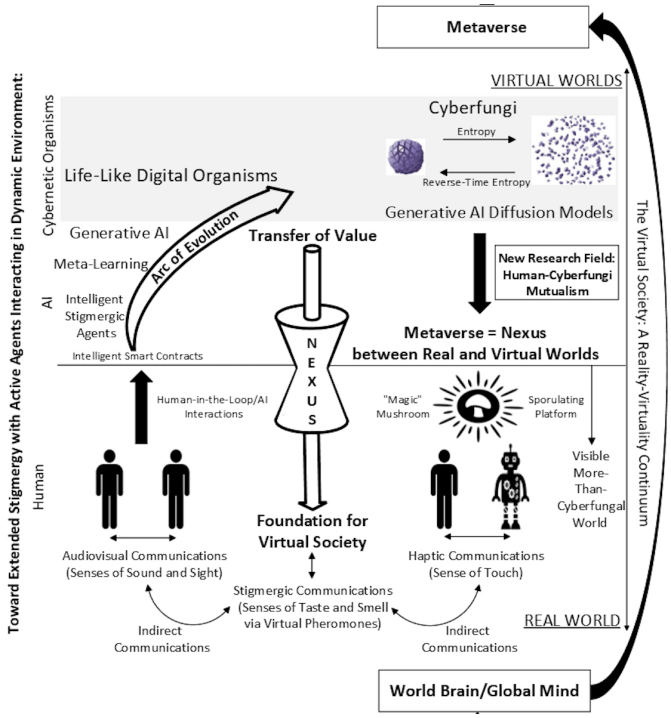

Fig. 2. 6G world brain in Metaverse comprises two complementary processes: (i) Left-hand side – Human-in-the-loop/AI interactions (upward arrow ▲) and (ii) Right-hand side – Embodied & enactive AGI (downward arrow ▼).

attention to facilitating advanced human-AI interaction capabilities. Humans may keep engaging in conventional types of multimodal communications, which come in two flavours: (i) human-to-human audiovisual communications (involving senses of sound and sight) and (ii) human-to-robot haptic communications (involving sense of touch). Besides these two types of direct human-to-human/robot communications, humans may also engage in indirect communications (i.e., stigmergic communications), where they leave digital pheromones in the virtual environment, similar to ants or other animals. More interestingly, humans interact with AI, ranging from intelligent smart contracts based on simple what-if programming to GenAI and more advanced cybernetic organisms (i.e., cyborgs), nicely resonating with Licklider's cyborg AI vision. These cybernetic organisms replicate life-like digital organisms via active inference, as explained in detail next.

Recall from Section III that biologically informed AI models play a central role in the emerging field of NeuroAI. For the biologically informed design of our life-like digital organism of choice, we are particularly interested in the so-called *botanical multiverse* of Peter Barlow, one of the world's most respected conceptual thinkers in plant neurobiology [20]. The botanical multiverse grows roots through the soil away from a host plant to interact with other roots via fungal networks, therefore forming an ancient mutualism between plants and *fungi*. These fungi are life-forms that perform 'active swarming,' which is a fundamental property of organisms they arrived at by an evolutionary convergence equipped with a capacity for cooperation that mutually benefits the different species of plants and fungi and their gene pool (for further details we refer the interested reader to [20]).

Typically, active inference models are built using the concept of *partially observable Markov decision process (POMDP)*, which offers a fairly expressive structure to model discrete state-space environments by means of parameters that can be expressed as tractable categorical distributions, including a set S of states s causing observations o and a set U of actions u. We developed an active inference model, called *cyberfungi* and shown on right-hand side of Fig. 2, which biomimics the following salient features of fungal networks:

- **Exploration via Branching:** Fungal networks *branch* out into soil in an exploratory way by relying on fungal connections, resulting in increased disorder (i.e., entropy). These fungal connections are realized by fungi that interconnect the roots of different trees.
- **Exploitation via Fusing:** Upon discovering resources, new fungal connections are *fused* that re-establish order (i.e., reverse-time entropy).
- **Embodiment via Entreating:** Fungi use mushrooms – fungi's fruiting bodies – to *entreat* the more-than-fungal world above ground. Mushrooms enable the embodiment of fungi to help them engage in agency.

Next, we demonstrate the superiority of our developed active inference model to a widely studied GenAI model known as denoising diffusion probabilistic model (DDPM). Figure 3 depicts the belief-update magnitude Δ_{KL}, which measures the KL divergence between the current and previous belief of a given agent at each network reconfiguration/rewiring step. Each belief-update corresponds to an "aha!" moment for the agent. We observe that DDPM exhibits only a single "aha!" moment early on, while further random rewiring steps produce minimal additional belief updates. By contrast, active inference continually guides the agent to make the most epistemically valuable structural changes throughout the entire lifetime of the rewiring process, resulting in a more intelligent lifelong learning process in active inference than GenAI DDPM.

As shown in Fig. 3, an interesting research question is the following: Is there a way to *enrich* the lifelong learning process

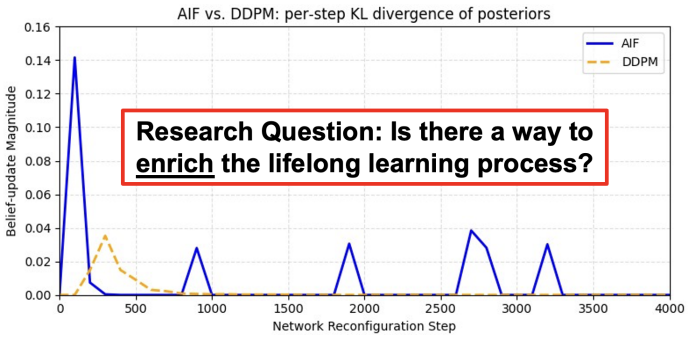

Fig. 3. Active inference (AIF) lifelong learning via "aha!" moments in contrast to GenAI DDPM: Mean belief-update magnitude Δ_{KL} obtained from 70 simulation runs for each measurement point.

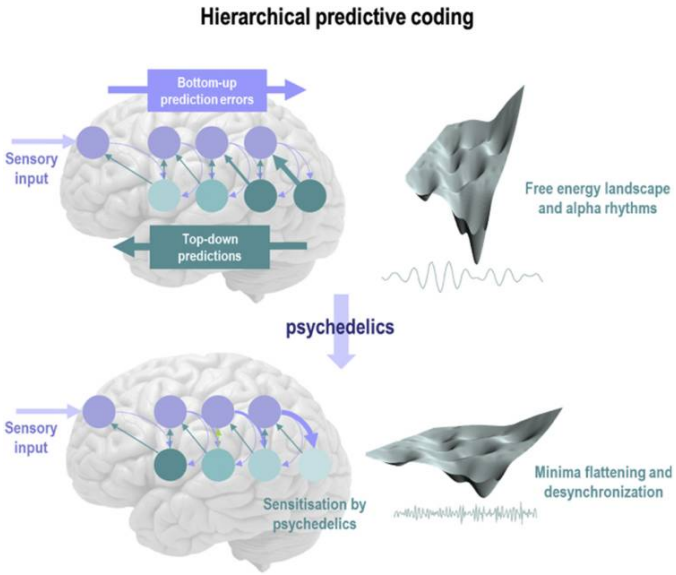

Fig. 4. Psychedelics have two positive effects on the brain: (i) expanding the free-energy landscape (i.e., opening up the mind), and (ii) reducing the curvature of the free-energy landscape (i.e., flattening local minima) [21].

for advancing active inference? Let's find out next.

C. Advancing Active Inference: From Biological Agents to Psychoactive Agents

In the mathematical framework of active inference, *entropy* is defined as the average surprise of observations. In [21], Friston *et al.* elaborated on the entropic effect of psychedelics on brain activity by *relaxing the precision of prior beliefs*, i.e., their ascribed importance or salience. They proposed a new brain model coined RElaxed Beliefs Under pSychedelics (REBUS) that integrates insights from the FEP with those of the so-called *entropic brain theory (EBT)*. Curiously, psychedelics appear to temporarily breach the FEP by increasing (rather than minimizing) brain entropy. The entropic brain effect of psychedelics increases the complexity of brain connectivity, leading to elevated brain entropy over an expanded, flattened free energy landscape, as illustrated in Fig. 4. Psychedelics promote a window of exceptionally high *neuroplasticity* that favors divergent thinking – a key component of creative thinking and discovering novelty that was previously hidden from clear sight. Insights of this sort are often felt as transcendent in their profundity. Interestingly, as the psychedelic effect begins to subside, the brain settles back into its default regimen of free energy minimization, but typically will not return entirely as before – a positive residual impact often referred to as the *psychedelic afterglow*. Clearly, the open research challenge is how to map the positive effects of psychoactives to specific parameters of the POMDP-based active inference model.

For illustration, Fig. 5 shows the evolving entropy of prior belief $\mathbf{D}$ during a 500-step long psychedelic window, where we mapped the weakened precision weighting of prior belief $\mathbf{D}$ to a decreased precision parameter ω of our developed POMDP-based active inference model. The resulting loss of

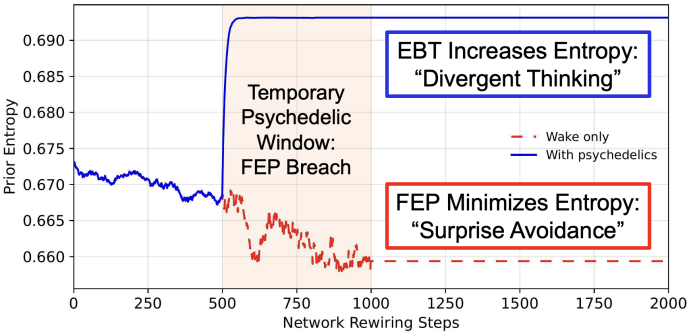

Fig. 5. Entropy of prior belief, captured by POMDP-based model parameter $\mathbf{D}$, under the temporary influence of psychedelics: Shaded region represents time window of psychedelic experience temporarily breaching FEP.

precision produces a free energy landscape with expanded state space accessibility. Note that Fig. 5 compares two agents, one with psychedelics and one without (wake only). While the awake agent experiences the prior entropy approaching its minimum (FEP regimen), the former one undergoes a sharp increase in prior entropy during the psychedelic window (breached FEP). After precision is restored (step 1000), the prior entropy remains elevated, showing a persistent effect resembling the psychedelic afterglow.

Note that Fig. 5 illustrates how to go beyond the FEP's overarching goal of surprise minimization (i.e., "surprise avoidance") by exploiting EBT to increase brain entropy (rather than minimize it), thus cracking open the door of human perception in the future 6G world brain by harnessing its enhanced neuroplasticity. Importantly, human-AI interaction with active inference agents enhanced by biomimetic psychoactives will help us humans penetrate ever deeper into cognitive space.

V. Active Inference 2.0: A Research Outlook

In the previous section, we have advanced active inference by widening its scope from biological agents to psychoactive agents. In this section, we provide a research outlook on the possible future of active inference by taking a look at the emerging field of *quantum biology*, where biological agents may be given an advantage through quantum effects.

Despite historically fierce criticism by both physicists and biologists, evidence suggests that nature has already beaten us in leveraging quantum effects to achieve something an equivalent classical system cannot, e.g., photosynthesis with an energy efficiency of up to 99% in bacteria and plants providing energy for almost all life on Earth. There is the possibility that there are many more examples of functional quantum behavior waiting to be discovered, raising several obvious broader questions, most notably: Can we learn from nature's examples and develop *biomimetic quantum technologies* [22]?

Originally put forth by quantum physicist Erwin Schrödinger in his book "What is Life?," quantum biology has become a modern field of research nowadays. In particular, the hypothesis that quantum mechanics may play a role in brain function, especially in the context of consciousness, has

recently sparked considerable interest and debate, giving rise to the term of *quantum consciousness* proposed by Roger Penrose (see discussion on artificial vs. natural intelligence above in Section III). Nevertheless, recent experimental and theoretical developments continue to keep the debate alive, such as the peculiar effects of anesthetics on making someone unconscious (as opposed to psychedelics that induce heightened states of consciousness). These insights illustrate how many effects, both classical and quantum, have to be taken into account when describing a biological process [23].

Quantum biology is the study of such biological processes. It is the application of quantum theory to aspects of biology for which classical physics fails to give an accurate description. Importantly, it asks the question whether there are essential *biological functions that appear classical but in reality are not?* The future role of quantum biology is precisely to expose and unravel this connection. More immediately, quantum biology promises to give rise to design principles for *biologically inspired quantum technologies*, which can be intrinsically useful independently from the organisms that inspired them. Through engineering such systems, it may be possible to test and quantify the extent to which quantum effects can enhance processes and functions found in biological agents [24].

Given that quantum biology provides a glimpse into the future of how we may have to rethink the ways in which energy and information are carried around our nervous system – thereby enabling us to reverse entropy for the sake of our survival – the future of active inference, which we shall call *active inference 2.0*, will delve below the surface of living organisms and make sense of what is going on beneath familiar activities such as thinking, where miniscule quantum entities are at work that play a vital role in human biology. Leveraging on active inference's inherent capability of creating a causal world model, active inference 2.0 may infer quantum causes from biological consequences and help us rethink processes which we have all been taken for granted. In fact, according to [25], this may help us gain a deeper understanding of how life on Earth originated with structures that could grow and reproduce, which is one of the oldest questions philosophers and scientists have asked themselves and it remains among the most difficult to answer.[2] The use of active inference 2.0 for biomimicking quantum effects in emerging AGI may be instrumental in eventually *reinventing the sacred*, as Stuart A. Kauffman prefers to call it (see also footnote in Section I).

VI. Conclusion

Similar to the (almost) forgotten human-centered AI vision of early-day Internet pioneers such as Engelbart and Licklider, scientists should also remember Albert Einstein's famous one-page op-ed in the New York Times, where Einstein convincingly argues that "the only deeply religious people of our largely materialistic age are the earnest men of research" [26]. This paper laid out our ideas on how to translate their visions from the past to the future shaped by the AGI-native 6G world brain premised on active inference, a cornerstone of NeuroAI.

[2]Note that the ultimate aim of xAI's Grok, the presumably most intelligent model in the world, is also the inquiry about the nature of reality by achieving a deep understanding of and interaction with it (visit https://x.ai for details).

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